Smart Vehicle Management System

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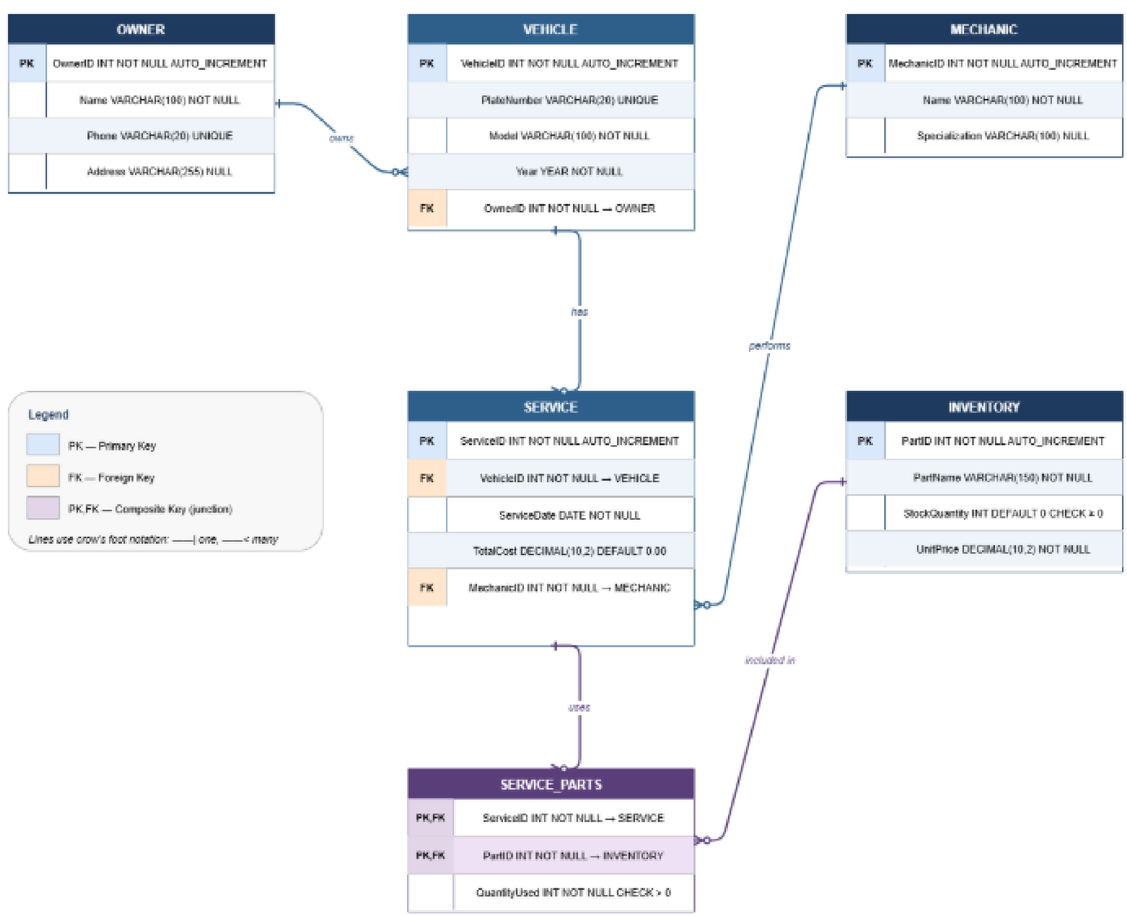
Milestone	Date	Version	Remarks
ERD Diagram and Schema	26-04-2026	1.0	

Part 1: Entity-Relationship Diagram (ERD)

The ERD below was created digitally using draw.io / mermaid-based tooling. It shows all six entities of the system, their attributes, primary keys (PK), foreign keys (FK), and the cardinality of every relationship.



Part 2: Relational Database Schema

The system consists of six relational tables. Each table is described below with its attributes, data types, key constraints, and relationships to other tables.

2.1 OWNER

Stores personal details of vehicle owners. Each owner can be linked to one or more vehicles.

Attribute	Data Type	Constraint	Description
OwnerID	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT	Unique identifier for each owner
Name	VARCHAR (100)	NOT NULL	Full name of the owner
Phone	VARCHAR (20)	NOT NULL, UNIQUE	Contact phone number
Address	VARCHAR (255)	NULL	Residential or business address

2.2 VEHICLE

Records every vehicle registered in the system. Each vehicle is associated with exactly one owner (foreign key to OWNER).

Attribute	Data Type	Constraint	Description
VehicleID	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT	Unique vehicle identifier
PlateNumber	VARCHAR (20)	NOT NULL, UNIQUE	Vehicle registration / plate number
Model	VARCHAR (100)	NOT NULL	Make and model of the vehicle
Year	YEAR	NOT NULL	Manufacturing year
OwnerID	INT	FOREIGN KEY → OWNER(OwnerID), NOT NULL	References the owning customer

2.3 MECHANIC

Stores details of mechanics employed at the garage. A mechanic can be assigned to multiple service records.

Attribute	Data Type	Constraint	Description
MechanicID	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT	Unique mechanic identifier
Name	VARCHAR (100)	NOT NULL	Full name of the mechanic
Specialization	VARCHAR (100)	NULL	Area of technical expertise (e.g., Engine, Electrical)

2.4 SERVICE

Logs every service event. Each service is tied to a specific vehicle and mechanic, and records the date and total cost billed.

Attribute	Data Type	Constraint	Description
ServiceID	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT	Unique service record identifier
VehicleID	INT	FOREIGN KEY → VEHICLE(VehicleID), NOT NULL	References the serviced vehicle
ServiceDate	DATE	NOT NULL	Date on which the service was performed
TotalCost	DECIMAL (10,2)	NOT NULL, DEFAULT 0.00	Total cost billed for the service
MechanicID	INT	FOREIGN KEY → MECHANIC(MechanicID), NOT NULL	References the assigned mechanic

2.5 INVENTORY

Tracks all spare parts held in stock. Stock levels are updated automatically when parts are used in a service.

Attribute	Data Type	Constraint	Description
PartID	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT	Unique spare part identifier
PartName	VARCHAR (150)	NOT NULL	Descriptive name of the part
StockQuantity	INT	NOT NULL, DEFAULT 0, CHECK (≥ 0)	Current quantity available in stock

Attribute	Data Type	Constraint	Description
UnitPrice	DECIMAL (10,2)	NOT NULL	Price per unit of the part

2.6 SERVICE_PARTS (Junction / Associative Table)

Resolves the many-to-many relationship between SERVICE and INVENTORY. Each row records how many units of a specific part were consumed during a specific service.

Attribute	Data Type	Constraint	Description
ServiceID	INT	PRIMARY KEY (composite), FOREIGN KEY → SERVICE(ServiceID)	References the service event
PartID	INT	PRIMARY KEY (composite), FOREIGN KEY → INVENTORY(PartID)	References the spare part used
QuantityUsed	INT	NOT NULL, CHECK (> 0)	Number of units of this part consumed

Part 3: Relationships Summary

Relationship	Cardinality	FK Location	Meaning
OWNER → VEHICLE	One-to-Many (1:M)	VEHICLE.OwnerID	One owner can own many vehicles; each vehicle has one owner
VEHICLE → SERVICE	One-to-Many (1:M)	SERVICE.VehicleID	One vehicle can have many service records; each service belongs to one vehicle
MECHANIC → SERVICE	One-to-Many (1:M)	SERVICE.MechanicID	One mechanic can perform many services; each service is assigned to one mechanic
SERVICE ↔ INVENTORY	Many-to-Many (M:N)	SERVICE_PARTS (junction)	A service can use many parts; a part can be used in many services

Part 4: Integrity Constraints

The following constraints enforce data consistency across all tables:

- **Primary Keys —** Every table has a surrogate INT primary key (AUTO_INCREMENT). SERVICE_PARTS uses a composite PK (ServiceID, PartID).
- **Foreign Keys —** All FK columns are enforced at the database level with ON DELETE RESTRICT / ON UPDATE CASCADE to preserve referential integrity.
- **NOT NULL —** Core identifiers (Name, PlateNumber, ServiceDate, TotalCost, PartName, UnitPrice) are mandatory.
- **UNIQUE —** VEHICLE.PlateNumber and OWNER.Phone are unique to prevent duplicate registrations.
- **CHECK —** INVENTORY.StockQuantity >= 0 prevents negative stock. SERVICE_PARTS.QuantityUsed > 0 ensures at least one unit is recorded.
- **Default Values —** TotalCost defaults to 0.00; StockQuantity defaults to 0 for newly added parts.